Supplementary Material

The Reach of Suction-Triggered Irrigation and the Marginal Fate of Irrigation Water: Closed Forms from the van Genuchten–Mualem Parameters

Derivations of the equations, the mechanisms behind the results, the data, and the origin of every table and figure

Naziru Halilu[1,2,*]

[1] Agricultural University of Athens, Athens, Greece

[2] Public University of Navarre, Pamplona, Spain

ORCID: 0009-0006-2324-9599

[*] Correspondence: halilunaziru73@gmail.com

Contents

S1 Highlights **3**

S2 What is new in this article **3**

S3 A step-by-step guide for students **4**

S3.1 Step 1: what a retention curve says 4

S3.2 Step 2: what the conductivity curve says 5

S3.3 Step 3: why a crop cannot dry a soil beyond a certain suction 5

S3.4 Step 4: a simple formula for the conductivity of a dry soil 6

S3.5 Step 5: solve for the reach 6

S3.6 Step 6: how κ was found and tested 6

S3.7 Step 7: where an irrigation millimetre goes 7

S3.8 Step 8: the dose law 7

S3.9 Step 9: when the sensor's lag matters 7

S3.10 Step 10: reproduce everything 8

S4 Derivations of the equations **8**

S4.1 The van Genuchten–Mualem pair from a capillary bundle 8

S4.2 The dry-end asymptote of the conductivity 9

S4.3 The reach of a trigger 10

S4.4 The alternative laws 10

S4.5 Fitting interval-censored reaches 11

S4.6 The marginal fate of irrigation 11

S4.7 The dose law 12

S4.8 Dynamic non-equilibrium and the lag of a sensor 12

S5 Why the results happen **13**

S6 The data **14**

S7 Where every table and figure comes from **28**

S1. Highlights

- A closed form from α, n and ℓ gives the largest suction a crop can impose at a sensor: the reach of a trigger.
- It predicts held-out soils within a factor of 2.3; triggers beyond it never fire and leave the crop rainfed.
- Of each irrigation millimetre, 38 % is transpired and 16 % drains; the rest evaporates.
- The drained share rises with the dose on a soil-independent scale of about 4 mm.
- Data (every log of every run), code, software and a 1155-page manual are open.

S2. What is new in this article

The table sets each contribution against what was established before it. The derivations that follow mark which equations are taken from the literature and which are new.

Table S1: What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

#	Established before	New in this article	Evidence here
1	Field capacity has intrinsic definitions from the hydraulic parameters (Assouline and Or, 2014; Twarakavi et al., 2009); uptake becomes limited by the conductance of the soil around the roots (Carminati and Javaux, 2020; Gardner, 1960).	A closed form for the dry-end counterpart, the *reach* of a suction trigger, with one constant shared by all soils.	Held-out soils within a factor of 2.3 (LOSO 0.18 decades); a threshold on absolute conductivity rejected.
2	Triggers of 300 to 600 cm are tabulated for field crops (Allen et al., 1998).	These triggers lie beyond the reach of coarse soils: they never fire and the crop stays rainfed.	Sand and loamy sand at 300 cm: rainfed yield; a deeper sensor does not help (E3).
3	–	A reach-scaled trigger h_t/h_c that places the irrigation responses of all soils on one axis.	Between-soil scatter falls from 54 % to 32 % of the total.

#	Established before	New in this article	Evidence here
4	Pulsing, discharge and hydraulic properties change losses below the root zone, shown case by case (Elmaloglou and Diamantopoulos, 2007, 2008, 2009).	The marginal fate of an irrigation millimetre across soils, doses, seasons and methods.	0.38 transpired, 0.25 soil evaporation, 0.19 other evaporation, 0.16 drained.
5	–	A dose scale of the drained share that does not depend on the soil, and a soil-specific plateau.	$d_D = 4.1 \pm 0.6$ mm, $\rho = 0.07$ with n; plateau 0.10 to 0.45.
6	Interception losses of sprinklers are known.	Overhead irrigation adds no transpiration: intercepted water replaces it.	$\eta_T = -0.08$ (all soils) and -0.01 (coarse loam).
7	Flow can be out of equilibrium with the retention curve (Diamantopoulos and Durner, 2012; Ross and Smettem, 2000).	A criterion, $\Pi_\tau = \tau / t_i$, for when this changes the firing of a trigger (proposed, not tested).	Derivation in Section S3.8.

S3. A step-by-step guide for students

This section explains, with numbers that can be checked on a calculator or in the workbook (sheets `EQ_P4_reach`, `EQ_P4_fate`, `EQ_P4_dose`), how every result of the article is obtained. Only school algebra, logarithms and the exponential function are needed. The full mathematical derivations follow in the next section.

S3.1 Step 1: what a retention curve says

A soil holds water in pores of many sizes. When the soil dries, the large pores empty first because water is held in them by weaker capillary forces. The *suction* h (in cm of water) measures how strongly the remaining water is held: 0 cm means saturated, 100 cm is about field capacity, 15 000

cm is about the wilting point. The *effective saturation* S_e is the fraction of the movable water still in the soil (1 = full, 0 = empty). The van Genuchten curve links the two:

$$S_e = \left[1 + (\alpha h)^n\right]^{-m}, \qquad m = 1 - 1/n.$$

α (1/cm) tells at which suction the soil starts to drain, n tells how sharply it drains: a sand has a large n (it drains almost all at once), a clay a small n (it drains slowly over a wide range). *Example (sand, $\alpha = 0.145$, $n = 2.68$).* $m = 1 - 1/2.68 = 0.627$. At $h = 100$ cm: $\alpha h = 14.5$; $14.5^{2.68} = 1\,293$; $1 + 1\,293 = 1\,294$; $1\,294^{-0.627} = 0.011$. The sand at field capacity holds only 1.1 % of its movable water.

S3.2 Step 2: what the conductivity curve says

Water moves through filled pores. As the soil dries, fewer and thinner pores carry water, so the *relative conductivity* K_r (1 = saturated) falls very fast. Mualem's formula gives

$$K_r = S_e^{1/2}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2.$$

Example (sand at 100 cm). $S_e = 0.0112$, $S_e^{1/m} = 0.0112^{1.595} = 7.7 \times 10^{-4}$; $1 - 7.7 \times 10^{-4} = 0.99923$; $0.99923^{0.627} = 0.99952$; $1 - 0.99952 = 4.8 \times 10^{-4}$; squared $= 2.3 \times 10^{-7}$; times $\sqrt{0.0112} = 0.106$ gives $K_r \approx 2.5 \times 10^{-8}$. At field capacity the sand already conducts 40 million times less than when saturated.

S3.3 Step 3: why a crop cannot dry a soil beyond a certain suction

Roots can only take water as fast as the soil brings it to them, and the soil brings water in proportion to its conductivity. Once K_r falls below a tiny value κ, the soil around the roots stops delivering water, the crop is stressed, and the soil at the sensor does not dry any further. The suction where this happens is the *reach* h_c of the soil. A trigger set drier than the reach is never read by the sensor, so the field is never irrigated.

S3.4 Step 4: a simple formula for the conductivity of a dry soil

When the soil is dry (αh much larger than 1), the 1 in the retention curve can be neglected, and the conductivity becomes a straight line on a log–log plot:

$$K_r \approx m^2(\alpha h)^{-E}, \qquad E = 2n + \tfrac{1}{2}(n-1).$$

Check (sand). $E = 2(2.68) + 0.5(1.68) = 6.2$; $m^2 = 0.393$; at $h = 100$ cm, $14.5^{-6.2} = 6.3 \times 10^{-8}$; $0.393 \times 6.3 \times 10^{-8} = 2.5 \times 10^{-8}$, the same as Step 2.

S3.5 Step 5: solve for the reach

Set $K_r(h_c) = \kappa$ in Step 4 and solve for h_c: take both sides to the power $-1/E$, then divide by α:

$$h_c = \frac{1}{\alpha}\left(\frac{m^2}{\kappa}\right)^{1/E}.$$

The constant κ is found once from the simulations (Step 6) and is the same for every soil: $\kappa = 3.47 \times 10^{-10}$. *Example (sand).* $m^2/\kappa = 0.393/3.47 \times 10^{-10} = 1.13 \times 10^9$. Its natural logarithm is 20.85; divided by $E = 6.2$ gives 3.363; $e^{3.363} = 28.9$; divided by $\alpha = 0.145$ gives $h_c = 199$ cm. *Example (loam, $\alpha = 0.036$, $n = 1.56$).* $m = 0.359$, $m^2 = 0.129$, $E = 3.4$; $\ln(0.129/3.47 \times 10^{-10}) = 19.73$; $19.73/3.4 = 5.80$; $e^{5.80} = 331$; $331/0.036 = 9\,209$ cm. So the sand's reach is 199 cm and the loam's is 9 209 cm: a 300 cm trigger works on loam but is never reached on sand.

S3.6 Step 6: how κ was found and tested

For each soil the simulations give an interval for the reach: the driest trigger that still irrigates (at least 5 mm a year) and the next drier trigger, which does not. For sand that is 100 to 150 cm. A value of κ is tried, the reach of every soil is computed with Step 5, and the distance (in powers of ten) between each prediction and its interval is squared and added; κ is changed until this sum is smallest. To test the formula fairly, each soil is left out in turn, κ is found from the other nine, and the left-out soil is predicted. The worst prediction was a factor of 2.3 from its interval. In the workbook, change κ in cell B2 of `EQ_P4_reach` and watch every reach and error change.

S3.7 Step 7: where an irrigation millimetre goes

Compare an irrigated run with the same soil and weather without irrigation (the rainfed baseline). The differences in each water flux, divided by the irrigation, tell what each millimetre became. *Example: one run on coarse loam* (dose 10 mm, April to October). Irrigation $I = 222.5$ mm a year. Transpiration rose from 147.6 to 223.0 mm, $+75.4$ mm, so $\eta_T = 75.4/222.5 = 0.34$. Drainage rose from 6.1 to 39.5 mm, so $\eta_D = 33.4/222.5 = 0.15$. Soil evaporation rose from 146.4 to 228.8 mm, so $\eta_{E_s} = 0.37$. Other evaporation (ET minus transpiration minus soil evaporation) rose from 172.8 to 193.3 mm, so $\eta_{E_o} = 0.09$. The four add to 0.95; the remaining 0.05 went to storage and runoff. *Pooling the 100 irrigated runs of that soil.* Add all transpiration (21 328 mm), subtract 100 times the rainfed value ($100 \times 147.6 = 14\,762$), and divide by all irrigation (17 277 mm): $\eta_T = 6\,567/17\,277 = 0.38$. The sheet `EQ_P4_fate` does exactly this with `SUMIFS` for every soil.

S3.8 Step 8: the dose law

A small dose is used by the crop and the surface on the same day and cannot reach the deep soil; a large dose can. The share that drains therefore starts near zero and grows towards a ceiling η_∞:

$$\eta_D(d) = \eta_\infty\big(1 - e^{-d/d_D}\big).$$

Example (loam, $\eta_\infty = 0.197$, $d_D = 3.9$ mm). For a 1 mm dose: $1/3.9 = 0.26$, $e^{-0.26} = 0.77$, $1 - 0.77 = 0.23$, times 0.197 gives 0.045. For 10 mm: $e^{-2.6} = 0.074$, so $0.197 \times 0.926 = 0.18$. For 30 mm: almost the ceiling, 0.197. The dose scale d_D was about 4 mm on every soil, close to one day of crop water use; the ceiling differs between soils (0.10 to 0.45).

S3.9 Step 9: when the sensor's lag matters

In fast wetting and drying the water content can lag behind the value of the retention curve by a time τ. If the schedule irrigates every t_i days, the lag matters only when $\Pi_\tau = \tau/t_i$ is not small. *Example:* $\tau = 1$ h and irrigation every 2 days gives $\Pi_\tau = 1/48 = 0.02$, negligible; pulses every 30 min give $\Pi_\tau = 2$, important.

S3.10 Step 10: reproduce everything

Open `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`: the yellow cells are the inputs, the green cells the formulas of Steps 1 to 8. Or run `Manuscripts/reproducibility/analysis/p4_reach.py`, which reads every run and writes every number and figure of the article.

S4. Derivations of the equations

Sections S3.1 and S3.2 re-derive the van Genuchten–Mualem pair and its dry-end asymptote, which are established mathematics (Mualem, 1976; van Genuchten, 1980). Everything from S3.3 on is new in this article: the reach of a trigger and its properties, the alternative laws and their test, the interval-censored fit, the marginal-fate identity applied across soils, doses, seasons and methods, the dose law, and the non-equilibrium criterion for triggers (the relaxation equation itself is established (Ross and Smettem, 2000)).

S4.1 The van Genuchten–Mualem pair from a capillary bundle

A soil is idealised as a bundle of capillaries whose radii are distributed so that the water-filled fraction at suction h is the effective saturation S_e. The capillary (Young–Laplace) law relates the radius r of the largest filled pore to the suction, $r = 2\sigma\cos\gamma/(\rho_w g h)$, so $h \propto 1/r$. Mualem (1976) wrote the relative conductivity of such a medium as the product of a tortuosity–connectivity term S_e^{ℓ} and the square of a normalised integral of the pore radius over the filled pores:

$$K_r(S_e) = S_e^{\ell}\left[\frac{\int_0^{S_e} h(x)^{-1}\,\mathrm{d}x}{\int_0^{1} h(x)^{-1}\,\mathrm{d}x}\right]^2. \tag{S1}$$

van Genuchten (1980) chose the retention function $S_e = [1+(\alpha h)^n]^{-m}$, whose inverse is

$$h(x) = \frac{1}{\alpha}\left(x^{-1/m}-1\right)^{1/n}. \tag{S2}$$

Substituting Eq. (S2) and $y = x^{1/m}$ ($x = y^m$, $\mathrm{d}x = m\,y^{m-1}\mathrm{d}y$):

$$\int_0^{S_e}\frac{\mathrm{d}x}{h(x)} = \alpha m\int_0^{S_e^{1/m}}\left(y^{-1}-1\right)^{-1/n} y^{m-1}\,\mathrm{d}y = \alpha m\int_0^{S_e^{1/m}} y^{\,m-1+1/n}(1-y)^{-1/n}\,\mathrm{d}y. \tag{S3}$$

With the restriction $m = 1 - 1/n$ the exponent of y vanishes and $(1-y)^{-1/n} = (1-y)^{m-1}$, so the integral is elementary:

$$\int_0^{S_e} \frac{\mathrm{d}x}{h(x)} = \alpha m \int_0^{S_e^{1/m}} (1-y)^{m-1}\mathrm{d}y = \alpha\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]. \tag{S4}$$

The denominator of Eq. (S1) is the same expression at $S_e = 1$, that is α, and the closed form of the paper follows:

$$K_r(S_e) = S_e^{\ell}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{S5}$$

This is why the restriction $m = 1 - 1/n$ is not a convenience: it is the condition under which the Mualem integral has a closed form.

S4.2 The dry-end asymptote of the conductivity

For $\alpha h \gg 1$, $(\alpha h)^n$ dominates the bracket of the retention function and

$$S_e = \big[1 + (\alpha h)^n\big]^{-m} = (\alpha h)^{-mn}\big[1 + (\alpha h)^{-n}\big]^{-m} = (\alpha h)^{-(n-1)}\Big[1 - m(\alpha h)^{-n} + O\big((\alpha h)^{-2n}\big)\Big], \tag{S6}$$

since $mn = n - 1$. Write $u = S_e^{1/m}$, which is small. The binomial expansion $(1-u)^m = 1 - mu - \frac{m(1-m)}{2}u^2 + O(u^3)$ gives

$$1 - (1-u)^m = mu\Big[1 + \tfrac{1-m}{2}u + O(u^2)\Big]. \tag{S7}$$

Inserting into Eq. (S5),

$$K_r = m^2\, S_e^{\ell + 2/m}\Big[1 + (1-m)S_e^{1/m} + \dots\Big]. \tag{S8}$$

The exponent is $(\ell + 2/m)(n-1) = \ell(n-1) + 2n$, because $(n-1)/m = n$. Hence

$$K_r(h) = m^2(\alpha h)^{-[2n + \ell(n-1)]}\Big[1 + \varepsilon(h)\Big], \qquad \varepsilon(h) = O\big((\alpha h)^{-n}\big) + O\big((\alpha h)^{-n}\big), \tag{S9}$$

where both correction terms are of the order $(\alpha h)^{-n}$ (the first from the retention bracket, the second from $S_e^{1/m} = (\alpha h)^{-n}[1 + \dots]$). For the soils of the paper the reach lies at αh_c between 29 and 1 280, where $(\alpha h_c)^{-n}$ is below 10^{-3}: the asymptote is exact to better than one per mille at the reach.

S4.3 The reach of a trigger

The physical condition. During a drying cycle the crop takes water from the layer that holds the sensor at a rate u limited by the conductance of the soil between the bulk soil and the root surface (Gardner, 1960). For a cylindrical root of radius r_0 in a soil cylinder of radius r_1, steady radial flow gives

$$u \simeq \frac{2\pi L_v}{\ln(r_1/r_0)} \int_{h_{\text{root}}}^{h} K(h')\,\mathrm{d}h' \;\approx\; \frac{2\pi L_v}{\ln(r_1/r_0)}\,K(h)\,\Delta h, \tag{S10}$$

with L_v the root length density and Δh the head drop to the root. Because K falls by many decades while Δh is bounded by the plant (the wilting point), u collapses once $K(h)$ is small enough, and the soil at the sensor stops drying within the time of a season. The suction where this happens is the reach h_{c}.

One constant for all soils. If the reach is reached at one value κ of the relative conductivity, $K_r(h_{\mathrm{c}}) = \kappa$, Eq. (S9) gives

$$m^2(\alpha h_{\mathrm{c}})^{-[2n+\ell(n-1)]} = \kappa \quad\Longrightarrow\quad \boxed{h_{\mathrm{c}} = \frac{1}{\alpha}\Big(\frac{m^2}{\kappa}\Big)^{1/[2n+\ell(n-1)]}} \tag{S11}$$

Properties. (i) α only rescales the reach: αh_{c} depends on n, ℓ and κ alone. (ii) The logarithmic sensitivity to n is

$$\frac{\partial \ln(\alpha h_{\mathrm{c}})}{\partial n} = \frac{2}{m\,n^2\,[2n+\ell(n-1)]} - \frac{(2+\ell)\ln(m^2/\kappa)}{[2n+\ell(n-1)]^2}, \tag{S12}$$

whose second term dominates (with $\kappa = 3.5\times10^{-10}$, $\ln(m^2/\kappa) \approx 18$ to 21), so the reach shortens steeply as n grows: from $\alpha h_{\mathrm{c}} \approx 1\,280$ at $n = 1.23$ to 29 at $n = 2.68$. (iii) A larger ℓ shortens the reach because the conductivity collapses faster. (iv) For $n \to 1$ the exponent tends to 2 and $m \to 0$, so the reach tends to $\alpha^{-1}(m^2/\kappa)^{1/2}$, which vanishes slowly: very broad pore-size distributions are still reachable far into the dry range, as observed for the silty clay loam.

S4.4 The alternative laws

Fixed saturation. If the reach were set by a water content, $S_e(h_{\mathrm{c}}) = S_c$, inverting the retention function gives $h_{\mathrm{c}} = \alpha^{-1}(S_c^{-1/m} - 1)^{1/n}$. It ignores the conductivity and predicts the coarse soils worst (Table 2 of the paper).

Power law in $n-1$. $\alpha h_{\mathrm{c}} = A(n-1)^{-B}$ is an empirical two-parameter law; it fits slightly better but has no derivation.

Absolute conductivity. $K(h_{\mathrm{c}}) = K_s K_r(h_{\mathrm{c}}) = q_c$ would give $h_{\mathrm{c}} = \alpha^{-1}(m^2 K_s/q_c)^{1/[2n+\ell(n-1)]}$. With literature K_s the conductivity at the observed reaches spans more than three decades, so a single q_c is rejected.

S4.5 Fitting interval-censored reaches

For each soil the data give an interval $[\,h_{\mathrm{lo}}, h_{\mathrm{hi}})$: h_{lo} is the driest trigger that still brings at least 5 mm of irrigation a year and h_{hi} the next drier trigger, or $+\infty$ when even the driest trigger fires. The loss of a prediction $\hat{h}$ is the squared distance in decades to the interval,

$$\mathcal{L}(\hat{h}) = \begin{cases} (\log_{10} h_{\mathrm{lo}} - \log_{10} \hat{h})^2, & \hat{h} < h_{\mathrm{lo}}, \\ 0, & h_{\mathrm{lo}} \le \hat{h} \le h_{\mathrm{hi}}, \\ (\log_{10} \hat{h} - \log_{10} h_{\mathrm{hi}})^2, & \hat{h} > h_{\mathrm{hi}}, \end{cases} \tag{S13}$$

summed over soils and minimised over $\log_{10}\kappa$ (Nelder–Mead). Leave-one-soil-out validation refits κ without the soil and evaluates $\mathcal{L}$ on it; the reported error is $\sqrt{\mathcal{L}}$ in decades and the worst factor is $10^{\max\sqrt{\mathcal{L}}}$.

S4.6 The marginal fate of irrigation

For a run with irrigation I and its rainfed twin (same soil, weather and crop), the 20-year water balances are

$$P + I = T + E_s + E_o + D + \Delta S + Q, \qquad P = T_0 + E_{s,0} + E_{o,0} + D_0 + \Delta S_0 + Q_0. \tag{S14}$$

Subtracting and dividing by I gives the identity of the paper,

$$1 = \underbrace{\frac{T - T_0}{I}}_{\eta_T} + \underbrace{\frac{E_s - E_{s,0}}{I}}_{\eta_{E_s}} + \underbrace{\frac{E_o - E_{o,0}}{I}}_{\eta_{E_o}} + \underbrace{\frac{D - D_0}{I}}_{\eta_D} + \underbrace{\frac{\Delta S - \Delta S_0 + Q - Q_0}{I}}_{\eta_S}. \tag{S15}$$

The shares are marginal, not average: a share can be negative (overhead irrigation lowers transpiration) or exceed the average loss fraction. Pooling a group of runs uses the ratio of sums,

$\eta_X = \sum(X - X_0)/\sum I$, which weights each run by its irrigation and is therefore the share of the group's total irrigation.

S4.7 The dose law

Let a dose d be applied when the root zone has a deficit. Part of the dose is used by the crop and the surface before it can move below the roots; the rest can drain. If the water that can be used before redistribution grows with the dose and saturates, and the drained share of the remainder is a soil property η_∞, the simplest form with the right limits is

$$\eta_D(d) = \eta_\infty\big(1 - e^{-d/d_D}\big), \qquad \eta_D \to \eta_\infty \frac{d}{d_D}\ (d \ll d_D), \qquad \eta_D \to \eta_\infty\ (d \gg d_D). \tag{S16}$$

The small-dose limit is linear because a vanishing dose cannot drain; the large-dose limit is the share of any extra water that the soil passes on. The dose scale d_D is the dose at which $1 - e^{-1} = 63\,\%$ of the plateau is reached. The form is phenomenological: it was chosen for its limits and tested against the data (coefficient of determination 0.92 to 0.96 on seven soils), not derived from the Richards equation.

S4.8 Dynamic non-equilibrium and the lag of a sensor

In the effective approach of Ross and Smettem (2000) and Diamantopoulos et al. (2012) the water content relaxes towards the equilibrium content of the retention curve with a time constant τ:

$$\frac{\partial\theta}{\partial t} = \frac{\theta_{\mathrm{eq}}(h) - \theta}{\tau}. \tag{S17}$$

During a drying phase in which θ_{eq} falls at a steady rate $r = -\mathrm{d}\theta_{\mathrm{eq}}/\mathrm{d}t$, the solution after the transient is $\theta = \theta_{\mathrm{eq}} + r\tau$, so the soil is wetter than equilibrium by $r\tau$, and a sensor that infers suction from the equilibrium curve under-reads the suction by

$$\delta h \simeq \frac{r\tau}{C(h)}, \qquad C(h) = -\frac{\mathrm{d}\theta_{\mathrm{eq}}}{\mathrm{d}h}. \tag{S18}$$

Over a schedule interval t_i the drying is rt_i, so the relative error in the time of firing is of order $\Pi_\tau = \tau/t_i$, the criterion of the paper. Where $C(h)$ is small (the steep dry limb of a sand), δh is large for the same $r\tau$, so non-equilibrium matters most where the reach is shortest.

S5. Why the results happen

Why a trigger has a reach. The crop can only dry the soil at the sensor as fast as the soil delivers water to the roots. That delivery scales with the conductivity, which falls by many orders of magnitude as the soil dries. In a soil with a narrow pore-size distribution (large n, sands) the conductivity collapses within a narrow band of suction just past the air entry, so the crop stops drying the soil there; whatever trigger is set beyond that band is never read. In a soil with a broad distribution (small n, silty clay loams) the conductivity fades slowly over many decades of suction and the crop can still dry the soil at very high suctions.

Why a deeper sensor does not help. Below the sensor depth of the experiments the root density is smaller, so the soil dries even less, and in sands the deeper soil is also wetted by drainage from above. A deeper sensor therefore reads lower suctions, not higher ones.

Why scaling by the reach collapses the curves only in part. The reach sets where irrigation ends, because it is fixed by the conductivity. How irrigation fades before the reach depends on how often the soil reaches a given suction in a season, which depends on the water held between field capacity and that suction, a retention property that differs from the conductivity property that sets the reach.

Why less than half of an irrigation millimetre is transpired. Transpiration is limited by the energy available to the canopy and by the canopy cover. Water applied to a wet surface or to a canopy that does not cover the ground evaporates from the soil, from the ponded surface and from intercepted water, using the same energy; water applied beyond the storage of the root zone drains.

Why overhead irrigation adds no transpiration. Water intercepted by the canopy evaporates with the energy that would otherwise drive transpiration, so the canopy transpires less while it is wet. The intercepted evaporation replaces transpiration rather than adding to it, and the marginal transpiration is nil or slightly negative.

Why the dose scale does not depend on the soil. The water that a dose can lose to drainage is the part that the crop and the surface do not use before it redistributes below the roots. The crop uses about one day of evapotranspiration in that time whatever the soil, so the dose scale is set by the atmosphere and the crop and not by the soil.

Why the plateau depends on the soil. Once a dose is large, the share that drains is the share of the excess that the soil passes on before the crop uses it, a race between drainage and uptake. Coarse sand passes water fast (high conductivity at field capacity), fine soils hold it, so the plateau is highest in coarse sand.

Why early-season irrigation is more productive. Early in the season the canopy is growing and its demand rises, so applied water is transpired; late in the season the cereal ripens and is harvested, and the same water is evaporated from bare soil or drains.

S6. The data

Table S2 lists the 17 experiments of the data folder and Fig. S1 shows their outputs. The tables that follow are the result files of the analysis, printed in full or up to 220 rows; every one is also a sheet of the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, and every run is in `Data/Comprehensive_Summary`.

Table S2: The data of this study: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; Y yield (Mg ha^{-1}), I irrigation, D drainage at 1 m and ET (mm yr^{-1}), N_L nitrate leached (kg N ha^{-1} yr^{-1}), t run time of Daisy (s). Every row is one file of the folder, linked.

Exp.	File	Runs	Valid	Factors varied	Y	I	D	ET	N_L	t
E0	`Baselines`	10	10	soil	2.19	0	55	418	158	231
E1	`Doses`	200	189	soil, rate, dose	5.13	198	98	564	79	195
E2	`TriggerPressure`	200	197	soil, trigger	4.70	169	80	559	90	139
E3	`SensorDepth`	200	200	soil, sensor depth	4.94	199	86	583	86	146
E4	`Methods`	100	96	soil, method	5.01	164	88	542	83	147
E5	`SHP Dynamics`	300	300	soil, method, tillage	4.13	152	43	579	84	189
E6	`Mediterranean Pollution`	210	193	soil, method, trigger, N rate	3.64	84	239	620	62	118
E7	`Mulching`	200	200	soil, mulch	5.29	183	96	555	74	161
E8	`Rain Intensity Moisture`	1000	1000	initial suction, rain scale	0.68	0	-1	282	-0	6
E9	`Soil Moisture Rain`	1000	987	soil, initial suction, rain scale	0.75	0	19	247	33	7
E10	`Amount Timing Soil`	1000	954	soil, rate, dose, window	4.42	139	76	532	97	146
E11	`Temp Evap Rain`	1000	1000	rain scale, warming, demand	3.81	324	18	754	66	151
E12	`Sensor Trigger Rain`	1000	1000	trigger, sensor depth, rain scale	5.88	260	35	670	32	155
E13	`N Water Soil`	1000	975	soil, method, rate, dose, trigger, N rate	4.00	162	83	547	79	207
E14	`Mulch Rain`	1000	1000	rain scale, mulch	6.15	235	43	634	25	173
E15	`Method Depth Amount`	1000	1000	method, rate, dose, root depth	6.02	217	61	619	37	157
E16	`Edge Cases`	770	740	soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand	2.17	344	185	600	93	224

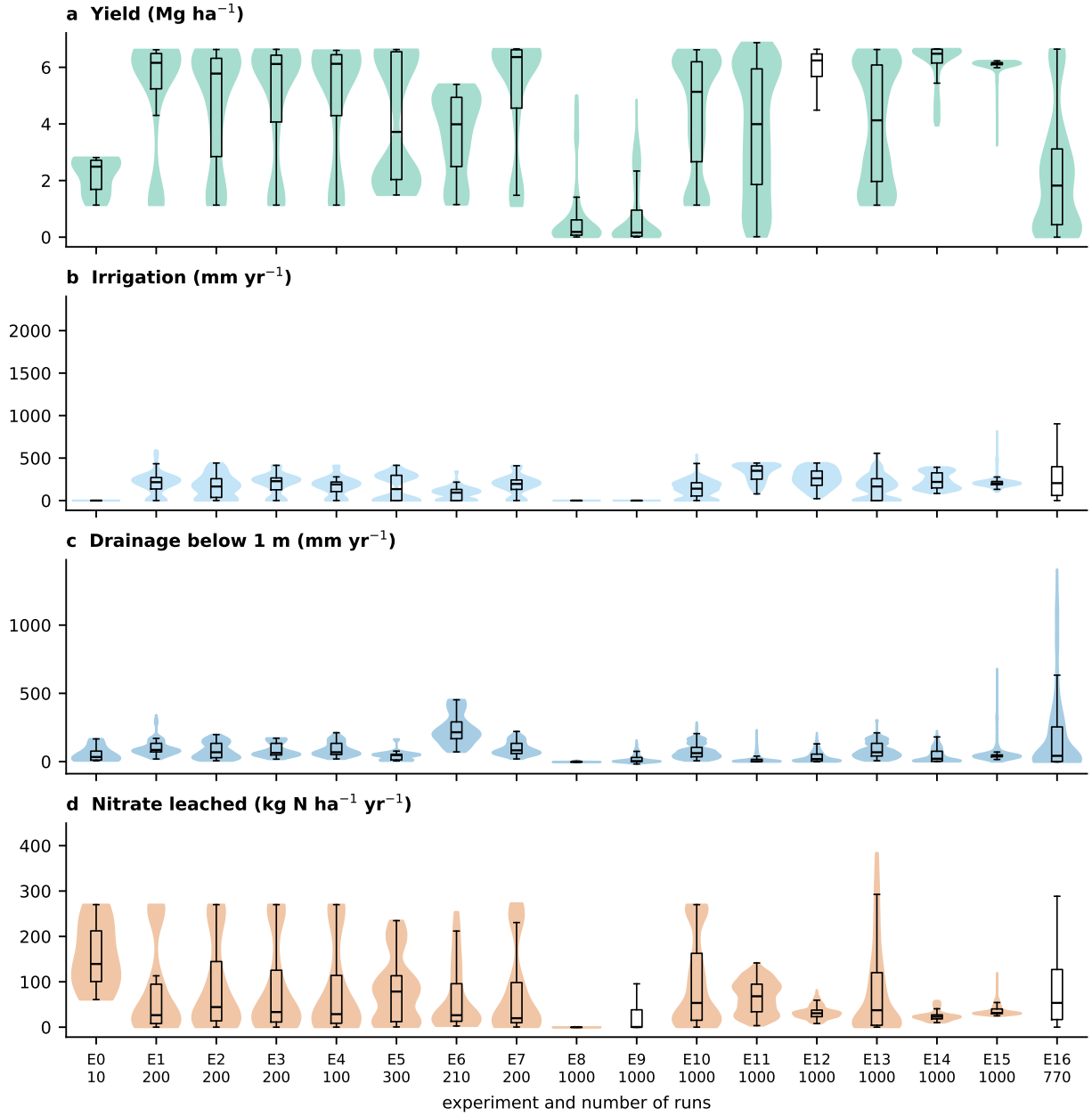


Figure S1: Yield, irrigation, drainage and nitrate leaching of the valid runs, by experiment.

p4_tableA1_reach.csv

Source: `Manuscripts/results/p4_tableA1_reach.csv`; 10 rows. Workbook sheet `R_p4_tableA1_reach`.

id	soil	alpha_1_per_cm	n	TAW_mm	reach_observed_low_cm	reach_observed_high_cm	reach_Eq4_cm	yield_at_300
S01	Coarse loam	0.0961	1.353	130	1e+04	inf	7837	6.158
S02	Coarse sand	0.0536	1.6	105	1e+04	inf	5373	6.254
S03	Sand	0.145	2.68	60	100	150	199	1.132
S04	Loamy sand	0.124	2.28	75	200	250	425.9	1.378
S05	Sandy loam	0.075	1.89	95	800	1000	1618	4.351
S06	Loam	0.036	1.56	150	5000	1e+04	9209	6.123
S07	Silt loam	0.02	1.41	220	1e+04	inf	2.961e+04	6.573

id	soil	alpha_1_per_cm	n	TAW_mm	reach_observed_low_cm	reach_observed_high_cm	reach_Eq4_cm	yield_at_300
S08	Sandy clay loam	0.059	1.48	160	5000	1e+04	7580	5.853
S09	Clay loam	0.019	1.31	175	1e+04	inf	4.776e+04	6.584
S10	Silty clay loam	0.01	1.23	195	1e+04	inf	1.283e+05	6.575

p4_table2_laws.csv

Source: `Manuscripts/results/p4_table2_laws.csv`; 3 rows. Workbook sheet `R_p4_table2_laws`.

law	theta	loso_rmse_log10	inside_of_10	worst_factor
mualem	-9.46	0.1753	5	2.3
power	1.832 3.12	0.1431	4	2.368
Sc	-1.489	0.2303	4	3.217

p4_fig4a_fate_by_soil.csv

Source: `Manuscripts/results/p4_fig4a_fate_by_soil.csv`; 8 rows. Workbook sheet `R_p4_fig4a_fate_by_soil`.

soil	transpired	soil_evaporation	other_evaporation	drained	storage_runoff
S01	0.3801	0.3228	0.1863	0.08046	0.03034
S02	0.2323	0.1865	0.2611	0.3161	0.00396
S05	0.6375	0.113	0.1369	0.07473	0.03789
S06	0.4093	0.2202	0.1739	0.1519	0.04473
S07	0.3523	0.2951	0.187	0.1478	0.01774
S08	0.4586	0.1843	0.1643	0.1474	0.04543
S09	0.3626	0.2869	0.1812	0.1529	0.01649
S10	0.3406	0.292	0.1772	0.1466	0.04357

p4_fig4b_fate_by_method.csv

Source: `Manuscripts/results/p4_fig4b_fate_by_method.csv`; 6 rows. Workbook sheet `R_p4_fig4b_fate_by_method`.

experiment	method	transpired	soil_evaporation	other_evaporation	drained
E4	overhead	-0.08	0.117	0.786	0.144
E4	subsoil	0.511	0.177	0.04	0.228
E4	surface	0.336	0.068	0.428	0.138
E15	overhead	-0.01	0.274	0.559	0.13
E15	subsoil	0.404	0.234	0.033	0.282
E15	surface	0.325	0.255	0.242	0.131

p4_window.csv

Source: `Manuscripts/results/p4_window.csv`; 10 rows. Workbook sheet `R_p4_window`.

window	etaT	etaD	I
4/1-10/1	0.344	0.209	1.85e+04
4/1-6/1	0.641	0.072	7917
4/1-7/1	0.477	0.153	1.262e+04
4/15-10/1	0.352	0.19	1.714e+04
5/1-10/1	0.348	0.164	1.592e+04
5/1-7/15	0.431	0.129	1.219e+04
5/15-10/1	0.34	0.157	1.538e+04
6/1-10/1	0.326	0.148	1.328e+04
6/1-8/15	0.362	0.119	1.097e+04
7/1-9/15	0.234	0.136	8153

p4_fig5_dose.csv

Source: `Manuscripts/results/p4_fig5_dose.csv`; 77 rows. Workbook sheet `R_p4_fig5_dose`.

soil	dose_mm	drained	transpired	evaporated	fit_plateau	fit_dose_scale_mm	fit_R2
S01	1	0.01116	0.2724	0.7167	0.1094	4.425	0.9492
S01	2	0.02367	0.3536	0.6117	0.1094	4.425	0.9492
S01	3	0.05371	0.3646	0.553	0.1094	4.425	0.9492
S01	4	0.07486	0.3772	0.5149	0.1094	4.425	0.9492
S01	6	0.08865	0.3887	0.4882	0.1094	4.425	0.9492
S01	8	0.09569	0.3917	0.4783	0.1094	4.425	0.9492
S01	10	0.09623	0.3973	0.4726	0.1094	4.425	0.9492
S01	15	0.1041	0.3972	0.464	0.1094	4.425	0.9492
S01	20	0.1051	0.4052	0.4561	0.1094	4.425	0.9492
S01	30	0.1088	0.3969	0.452	0.1094	4.425	0.9492
S02	1	0.09002	0.1999	0.707	0.4448	3.234	0.9548
S02	2	0.1722	0.2894	0.5354	0.4448	3.234	0.9548
S02	3	0.2796	0.2777	0.4387	0.4448	3.234	0.9548
S02	4	0.3611	0.2207	0.4135	0.4448	3.234	0.9548
S02	6	0.3737	0.2353	0.3861	0.4448	3.234	0.9548
S02	8	0.4047	0.182	0.41	0.4448	3.234	0.9548
S02	10	0.4119	0.1804	0.4044	0.4448	3.234	0.9548
S05	1	0.02737	0.3501	0.6058	0.1017	5.21	0.6575
S05	2	0.04118	0.5204	0.4112	0.1017	5.21	0.6575
S05	3	0.05739	0.5942	0.3121	0.1017	5.21	0.6575
S05	4	0.06418	0.6268	0.2689	0.1017	5.21	0.6575
S05	6	0.07016	0.6572	0.2316	0.1017	5.21	0.6575
S05	8	0.06801	0.6813	0.2103	0.1017	5.21	0.6575
S05	10	0.07391	0.6866	0.1991	0.1017	5.21	0.6575
S05	15	0.06929	0.7145	0.1781	0.1017	5.21	0.6575

soil	dose_mm	drained	transpired	evaporated	fit_plateau	fit_dose_scale_mm	fit_R2
S05	20	0.09447	0.695	0.1704	0.1017	5.21	0.6575
S05	30	0.1362	0.6673	0.1531	0.1017	5.21	0.6575
S06	1	0.01796	0.2696	0.7016	0.1971	3.887	0.9311
S06	2	0.06301	0.3512	0.5581	0.1971	3.887	0.9311
S06	3	0.1161	0.3796	0.4601	0.1971	3.887	0.9311
S06	4	0.1462	0.4004	0.4051	0.1971	3.887	0.9311
S06	6	0.165	0.4219	0.3629	0.1971	3.887	0.9311
S06	8	0.1693	0.432	0.3484	0.1971	3.887	0.9311
S06	10	0.1729	0.4367	0.3402	0.1971	3.887	0.9311
S06	15	0.1805	0.4459	0.3234	0.1971	3.887	0.9311
S06	20	0.1857	0.4441	0.3205	0.1971	3.887	0.9311
S06	30	0.2169	0.4317	0.3053	0.1971	3.887	0.9311
S07	1	0.01468	0.2724	0.7147	0.194	4.106	0.9292
S07	2	0.05912	0.3413	0.5963	0.194	4.106	0.9292
S07	3	0.1092	0.3436	0.5296	0.194	4.106	0.9292
S07	4	0.1408	0.3477	0.4905	0.194	4.106	0.9292
S07	6	0.1589	0.3575	0.4622	0.194	4.106	0.9292
S07	8	0.1641	0.3623	0.4525	0.194	4.106	0.9292
S07	10	0.1699	0.3642	0.445	0.194	4.106	0.9292
S07	15	0.174	0.3671	0.4382	0.194	4.106	0.9292
S07	20	0.1823	0.366	0.4307	0.194	4.106	0.9292
S07	30	0.2145	0.3564	0.4098	0.194	4.106	0.9292
S08	1	0.0296	0.2739	0.6874	0.19	3.85	0.9452
S08	2	0.06499	0.3832	0.5247	0.19	3.85	0.9452
S08	3	0.1097	0.4284	0.4178	0.19	3.85	0.9452
S08	4	0.1397	0.4525	0.3585	0.19	3.85	0.9452
S08	6	0.1563	0.4748	0.3173	0.19	3.85	0.9452
S08	8	0.1613	0.4865	0.3006	0.19	3.85	0.9452
S08	10	0.1666	0.4892	0.2926	0.19	3.85	0.9452
S08	15	0.1741	0.4977	0.2767	0.19	3.85	0.9452
S08	20	0.1798	0.5015	0.2683	0.19	3.85	0.9452
S08	30	0.2122	0.4892	0.2521	0.19	3.85	0.9452
S09	1	0.01645	0.2722	0.7152	0.2043	4.388	0.926
S09	2	0.06101	0.3491	0.5921	0.2043	4.388	0.926
S09	3	0.1096	0.3522	0.523	0.2043	4.388	0.926
S09	4	0.1438	0.3592	0.4765	0.2043	4.388	0.926
S09	6	0.1617	0.3707	0.4464	0.2043	4.388	0.926
S09	8	0.1695	0.3732	0.4365	0.2043	4.388	0.926
S09	10	0.173	0.3759	0.43	0.2043	4.388	0.926
S09	15	0.1765	0.3798	0.4228	0.2043	4.388	0.926
S09	20	0.1938	0.3783	0.4081	0.2043	4.388	0.926
S09	30	0.2299	0.3654	0.387	0.2043	4.388	0.926
S10	1	0.01687	0.2859	0.7031	0.1827	3.394	0.923
S10	2	0.06202	0.3556	0.5896	0.1827	3.394	0.923
S10	3	0.1119	0.3491	0.528	0.1827	3.394	0.923
S10	4	0.1439	0.3487	0.4899	0.1827	3.394	0.923
S10	6	0.1635	0.3542	0.4635	0.1827	3.394	0.923
S10	8	0.1683	0.3576	0.4561	0.1827	3.394	0.923

soil	dose_mm	drained	transpired	evaporated	fit_plateau	fit_dose_scale_mm	fit_R2
S10	10	0.1773	0.3561	0.4482	0.1827	3.394	0.923
S10	15	0.176	0.3599	0.4442	0.1827	3.394	0.923
S10	20	0.1911	0.3542	0.427	0.1827	3.394	0.923
S10	30	0.1629	0.2778	0.3331	0.1827	3.394	0.923

p4_fig3_trigger_runs.csv

Source: `Manuscripts/results/p4_fig3_trigger_runs.csv`; 197 rows. Workbook sheet `R_p4_fig3_trigger_runs`.

soil	trigger_cm	irrigation_mm	yield_Mg_ha	trigger_over_reach	Inorm
S01_CoarseLoam	100	296.9	6.62	0.01276	0.8858
S01_CoarseLoam	1000	168.7	5.928	0.1276	0.5034
S01_CoarseLoam	1e+04	60.48	3.798	1.276	0.1805
S01_CoarseLoam	1200	160.8	5.828	0.1531	0.4798
S01_CoarseLoam	150	270.8	6.317	0.01914	0.8081
S01_CoarseLoam	1500	152.3	5.707	0.1914	0.4544
S01_CoarseLoam	200	254	6.271	0.02552	0.758
S01_CoarseLoam	2000	142.1	5.526	0.2552	0.4239
S01_CoarseLoam	250	242.5	6.318	0.0319	0.7236
S01_CoarseLoam	300	228.5	6.158	0.03828	0.6817
S01_CoarseLoam	3000	124.1	5.087	0.3828	0.3702
S01_CoarseLoam	350	219	6.088	0.04466	0.6534
S01_CoarseLoam	400	211.8	6.078	0.05104	0.6319
S01_CoarseLoam	50	335.2	6.62	0.00638	1
S01_CoarseLoam	500	202.3	6.064	0.0638	0.6037
S01_CoarseLoam	5000	97.92	4.547	0.638	0.2922
S01_CoarseLoam	600	191.5	6.041	0.07656	0.5714
S01_CoarseLoam	700	183.8	6.022	0.08932	0.5485
S01_CoarseLoam	75	311.8	6.624	0.00957	0.9302
S01_CoarseLoam	800	177.5	5.996	0.1021	0.5295
S02_CoarseSand	100	438.8	6.317	0.01861	0.9938
S02_CoarseSand	1e+04	139.3	3.853	1.861	0.3155
S02_CoarseSand	150	433.2	6.306	0.02792	0.981
S02_CoarseSand	200	427.1	6.295	0.03723	0.9671
S02_CoarseSand	2000	248	4.861	0.3723	0.5617
S02_CoarseSand	250	419.4	6.281	0.04653	0.9497
S02_CoarseSand	300	411.1	6.254	0.05584	0.931
S02_CoarseSand	3000	218.5	4.586	0.5584	0.4948
S02_CoarseSand	350	402.6	6.222	0.06514	0.9117
S02_CoarseSand	400	394.1	6.189	0.07445	0.8924
S02_CoarseSand	50	441.6	6.32	0.009306	1
S02_CoarseSand	500	379.2	6.106	0.09306	0.8587
S02_CoarseSand	5000	180.7	4.283	0.9306	0.4092
S02_CoarseSand	600	366.4	6.044	0.1117	0.8296

soil	trigger_cm	irrigation_mm	yield_Mg_ha	trigger_over_reach	Inorm
S02_CoarseSand	700	352.4	5.916	0.1303	0.7981
S02_CoarseSand	75	441.2	6.32	0.01396	0.9992
S02_CoarseSand	800	341	5.844	0.1489	0.7723
S03_Sand	100	6.36	1.234	0.5024	0.05206
S03_Sand	1000	0	1.132	5.024	0
S03_Sand	1e+04	0	1.132	50.24	0
S03_Sand	1200	0	1.132	6.029	0
S03_Sand	150	0	1.132	0.7536	0
S03_Sand	1500	0	1.132	7.536	0
S03_Sand	200	0	1.132	1.005	0
S03_Sand	2000	0	1.132	10.05	0
S03_Sand	250	0	1.132	1.256	0
S03_Sand	300	0	1.132	1.507	0
S03_Sand	3000	0	1.132	15.07	0
S03_Sand	350	0	1.132	1.758	0
S03_Sand	400	0	1.132	2.01	0
S03_Sand	50	122.2	3.038	0.2512	1
S03_Sand	500	0	1.132	2.512	0
S03_Sand	5000	0	1.132	25.12	0
S03_Sand	600	0	1.132	3.014	0
S03_Sand	700	0	1.132	3.517	0
S03_Sand	75	38.52	1.765	0.3768	0.3153
S03_Sand	800	0	1.132	4.019	0
S04_LoamySand	100	122.9	3.455	0.2348	0.5508
S04_LoamySand	1000	0	1.378	2.348	0
S04_LoamySand	1e+04	0	1.378	23.48	0
S04_LoamySand	1200	0	1.378	2.818	0
S04_LoamySand	150	50.04	2.276	0.3522	0.2243
S04_LoamySand	1500	0	1.378	3.522	0
S04_LoamySand	200	16.8	1.67	0.4696	0.07531
S04_LoamySand	2000	0	1.378	4.696	0
S04_LoamySand	250	4.08	1.433	0.587	0.01829
S04_LoamySand	300	0	1.378	0.7044	0
S04_LoamySand	3000	0	1.378	7.044	0
S04_LoamySand	350	0	1.378	0.8218	0
S04_LoamySand	400	0	1.378	0.9392	0
S04_LoamySand	50	223.1	4.544	0.1174	1
S04_LoamySand	500	0	1.378	1.174	0
S04_LoamySand	5000	0	1.378	11.74	0
S04_LoamySand	600	0	1.378	1.409	0
S04_LoamySand	700	0	1.378	1.644	0
S04_LoamySand	75	152.8	3.893	0.1761	0.6848
S04_LoamySand	800	0	1.378	1.878	0
S05_SandyLoam	100	230	5.263	0.06179	0.6291
S05_SandyLoam	1000	4.08	1.986	0.6179	0.01116
S05_SandyLoam	1e+04	0	1.851	6.179	0
S05_SandyLoam	1200	0	1.851	0.7415	0
S05_SandyLoam	150	181.7	5.016	0.09268	0.4969

soil	trigger_cm	irrigation_mm	yield_Mg_ha	trigger_over_reach	Inorm
S05_SandyLoam	1500	0	1.851	0.9268	0
S05_SandyLoam	200	159.4	4.783	0.1236	0.4358
S05_SandyLoam	2000	0	1.851	1.236	0
S05_SandyLoam	250	143.3	4.579	0.1545	0.3919
S05_SandyLoam	300	127.6	4.351	0.1854	0.3489
S05_SandyLoam	3000	0	1.851	1.854	0
S05_SandyLoam	350	117.1	4.094	0.2163	0.3203
S05_SandyLoam	400	99.12	3.834	0.2472	0.2711
S05_SandyLoam	50	365.6	6.02	0.03089	1
S05_SandyLoam	500	72.12	3.314	0.3089	0.1972
S05_SandyLoam	5000	0	1.851	3.089	0
S05_SandyLoam	600	47.76	2.847	0.3707	0.1306
S05_SandyLoam	700	35.88	2.652	0.4325	0.09813
S05_SandyLoam	75	282.7	5.756	0.04634	0.7732
S05_SandyLoam	800	17.52	2.28	0.4943	0.04792
S06_Loam	100	346.4	6.606	0.01086	0.8178
S06_Loam	1000	116.2	5.189	0.1086	0.2742
S06_Loam	1e+04	0	2.615	1.086	0
S06_Loam	1200	104.6	4.909	0.1303	0.247
S06_Loam	150	294	6.552	0.01629	0.6941
S06_Loam	1500	87.12	4.597	0.1629	0.2057
S06_Loam	200	257.3	6.385	0.02172	0.6074
S06_Loam	2000	67.8	4.199	0.2172	0.1601
S06_Loam	250	230.3	6.161	0.02715	0.5436
S06_Loam	300	211.7	6.123	0.03258	0.4997
S06_Loam	3000	36.72	3.574	0.3258	0.08669
S06_Loam	350	201	6.098	0.038	0.4745
S06_Loam	400	189.6	6.063	0.04343	0.4476
S06_Loam	50	423.6	6.623	0.005429	1
S06_Loam	500	173	5.955	0.05429	0.4085
S06_Loam	5000	5.16	2.764	0.5429	0.01218
S06_Loam	600	158.8	5.806	0.06515	0.3748
S06_Loam	700	147.1	5.672	0.07601	0.3473
S06_Loam	75	381.4	6.62	0.008144	0.9003
S06_Loam	800	137.4	5.523	0.08687	0.3244
S07_SiltLoam	100	379.1	6.624	0.003377	0.8797
S07_SiltLoam	1000	166.9	6.198	0.03377	0.3874
S07_SiltLoam	1e+04	19.2	3.341	0.3377	0.04456
S07_SiltLoam	1200	157	6.06	0.04052	0.3642
S07_SiltLoam	150	338.5	6.622	0.005065	0.7856
S07_SiltLoam	1500	145.4	5.838	0.05065	0.3375
S07_SiltLoam	200	311.6	6.62	0.006754	0.7232
S07_SiltLoam	2000	125.5	5.529	0.06754	0.2913
S07_SiltLoam	250	290.6	6.628	0.008442	0.6745
S07_SiltLoam	300	271.8	6.573	0.01013	0.6307
S07_SiltLoam	3000	107.8	5.178	0.1013	0.2501
S07_SiltLoam	350	257.5	6.564	0.01182	0.5976
S07_SiltLoam	400	242.8	6.425	0.01351	0.5634

soil	trigger_cm	irrigation_mm	yield_Mg_ha	trigger_over_reach	Inorm
S07_SiltLoam	50	430.9	6.626	0.001688	1
S07_SiltLoam	500	218.6	6.388	0.01688	0.5074
S07_SiltLoam	5000	79.68	4.654	0.1688	0.1849
S07_SiltLoam	600	199.9	6.359	0.02026	0.4639
S07_SiltLoam	700	188.4	6.31	0.02364	0.4372
S07_SiltLoam	75	406.2	6.623	0.002533	0.9426
S07_SiltLoam	800	180.6	6.288	0.02702	0.4191
S08_SandyClayLoam	100	294.5	6.449	0.01319	0.7805
S08_SandyClayLoam	1000	109.3	4.773	0.1319	0.2898
S08_SandyClayLoam	1e+04	0	2.384	1.319	0
S08_SandyClayLoam	1200	97.2	4.527	0.1583	0.2576
S08_SandyClayLoam	150	247.2	6.144	0.01979	0.6552
S08_SandyClayLoam	1500	82.2	4.276	0.1979	0.2179
S08_SandyClayLoam	200	220.2	5.998	0.02639	0.5837
S08_SandyClayLoam	2000	64.2	3.923	0.2639	0.1702
S08_SandyClayLoam	250	205.4	5.91	0.03298	0.5445
S08_SandyClayLoam	300	191.5	5.853	0.03958	0.5076
S08_SandyClayLoam	3000	36.84	3.348	0.3958	0.09765
S08_SandyClayLoam	350	180.5	5.79	0.04617	0.4784
S08_SandyClayLoam	400	171.2	5.725	0.05277	0.4539
S08_SandyClayLoam	50	377.3	6.566	0.006596	1
S08_SandyClayLoam	500	158.5	5.554	0.06596	0.4202
S08_SandyClayLoam	5000	6	2.546	0.6596	0.0159
S08_SandyClayLoam	600	146.5	5.399	0.07916	0.3884
S08_SandyClayLoam	700	134.3	5.229	0.09235	0.3559
S08_SandyClayLoam	75	330.2	6.515	0.009895	0.8753
S08_SandyClayLoam	800	121.3	5.039	0.1055	0.3216
S09_ClayLoam	100	354.8	6.617	0.002094	0.8718
S09_ClayLoam	1000	174.5	6.311	0.02094	0.4287
S09_ClayLoam	1e+04	60.72	4.275	0.2094	0.1492
S09_ClayLoam	1200	165.5	6.21	0.02513	0.4065
S09_ClayLoam	150	324.1	6.618	0.003141	0.7963
S09_ClayLoam	1500	157.7	6.031	0.03141	0.3874
S09_ClayLoam	200	300.6	6.62	0.004188	0.7385
S09_ClayLoam	2000	143.6	5.778	0.04188	0.3529
S09_ClayLoam	250	282.5	6.622	0.005235	0.694
S09_ClayLoam	300	266.9	6.584	0.006282	0.6557
S09_ClayLoam	3000	124.3	5.436	0.06282	0.3054
S09_ClayLoam	350	253.7	6.55	0.007329	0.6232
S09_ClayLoam	400	241.6	6.474	0.008376	0.5935
S09_ClayLoam	50	407	6.605	0.001047	1
S09_ClayLoam	500	221.9	6.451	0.01047	0.5451
S09_ClayLoam	5000	105.5	5.079	0.1047	0.2591
S09_ClayLoam	600	205.2	6.427	0.01256	0.5041
S09_ClayLoam	700	193.9	6.409	0.01466	0.4764
S09_ClayLoam	75	377.6	6.62	0.00157	0.9278
S09_ClayLoam	800	185.8	6.386	0.01675	0.4564
S10_SiltyClayLoam	100	346.8	6.229	0.0007797	0.9051

soil	trigger_cm	irrigation_mm	yield_Mg_ha	trigger_over_reach	Inorm
S10_SiltyClayLoam	1000	190.3	6.425	0.007797	0.4967
S10_SiltyClayLoam	1e+04	95.04	4.905	0.07797	0.248
S10_SiltyClayLoam	1200	182	6.393	0.009356	0.4751
S10_SiltyClayLoam	150	324.6	6.418	0.00117	0.8472
S10_SiltyClayLoam	1500	175.1	6.318	0.0117	0.4569
S10_SiltyClayLoam	200	306	6.509	0.001559	0.7986
S10_SiltyClayLoam	2000	163.1	6.163	0.01559	0.4256
S10_SiltyClayLoam	250	291.5	6.547	0.001949	0.7607
S10_SiltyClayLoam	300	278.4	6.575	0.002339	0.7266
S10_SiltyClayLoam	3000	149.9	5.874	0.02339	0.3912
S10_SiltyClayLoam	350	268.2	6.585	0.002729	0.7
S10_SiltyClayLoam	400	259.8	6.59	0.003119	0.678
S10_SiltyClayLoam	50	383.2	5.986	0.0003898	1
S10_SiltyClayLoam	500	241.6	6.507	0.003898	0.6304
S10_SiltyClayLoam	5000	127.9	5.496	0.03898	0.3339
S10_SiltyClayLoam	600	225.4	6.491	0.004678	0.5882
S10_SiltyClayLoam	700	212.6	6.469	0.005458	0.555
S10_SiltyClayLoam	75	362.6	6.103	0.0005848	0.9464
S10_SiltyClayLoam	800	203.4	6.455	0.006237	0.5308

p4_numbers.json

Source: `Manuscripts/results/p4_numbers.json`; 295 values, first 220 shown. Workbook sheet `R_p4_numbers`.

quantity	value
runs_total	10190
runs_valid	10041
E2_runs	197
reach.mualem.theta	-9.46
reach.mualem.loso_err_log10	0.169, 0.362, 0.134, 0.26, 0.25, 0, 0, 0, 0, 0
reach.mualem.loso_rms_log10	0.1753
reach.mualem.loso_inside	5
reach.mualem.loso_max_factor	2.3
reach.mualem.pred.S01	7837
reach.mualem.pred.S02	5373
reach.mualem.pred.S03	199
reach.mualem.pred.S04	425.9
reach.mualem.pred.S05	1618
reach.mualem.pred.S06	9209
reach.mualem.pred.S07	2.961e+04
reach.mualem.pred.S08	7580
reach.mualem.pred.S09	4.776e+04
reach.mualem.pred.S10	1.283e+05
reach.power.theta	1.832, 3.12

quantity	value
reach.power.loso_err_log10	0, 0.374, 0.172, 0.009, 0.139, 0.086, 0, 0.092, 0, 0
reach.power.loso_rms_log10	0.1431
reach.power.loso_inside	4
reach.power.loso_max_factor	2.368
reach.power.pred.S01	1.82e+04
reach.power.pred.S02	6235
reach.power.pred.S03	92.83
reach.power.pred.S04	253.5
reach.power.pred.S05	1302
reach.power.pred.S06	1.151e+04
reach.power.pred.S07	5.481e+04
reach.power.pred.S08	1.136e+04
reach.power.pred.S09	1.38e+05
reach.power.pred.S10	6.654e+05
reach.Sc.theta	-1.489
reach.Sc.loso_err_log10	0, 0.317, 0.284, 0.243, 0.117, 0.138, 0, 0.507, 0, 0
reach.Sc.loso_rms_log10	0.2303
reach.Sc.loso_inside	4
reach.Sc.loso_max_factor	3.217
reach.Sc.pred.S01	1.723e+05
reach.Sc.pred.S02	5662
reach.Sc.pred.S03	53.02
reach.Sc.pred.S04	117.4
reach.Sc.pred.S05	628.3
reach.Sc.pred.S06	1.268e+04
reach.Sc.pred.S07	2.145e+05
reach.Sc.pred.S08	2.147e+04
reach.Sc.pred.S09	3.353e+06
reach.Sc.pred.S10	2.987e+08
kappa	3.47e-10
blind_300	S03, S04
yield_E2.S01.rainfed	2.597
yield_E2.S01.best	6.624
yield_E2.S01.at300	6.158
yield_E2.S02.rainfed	1.633
yield_E2.S02.best	6.32
yield_E2.S02.at300	6.254
yield_E2.S03.rainfed	1.132
yield_E2.S03.best	3.038
yield_E2.S03.at300	1.132
yield_E2.S04.rainfed	1.378
yield_E2.S04.best	4.544
yield_E2.S04.at300	1.378
yield_E2.S05.rainfed	1.851
yield_E2.S05.best	6.02
yield_E2.S05.at300	4.351
yield_E2.S06.rainfed	2.615
yield_E2.S06.best	6.623

quantity	value
yield_E2.S06.at300	6.123
yield_E2.S07.rainfed	2.813
yield_E2.S07.best	6.628
yield_E2.S07.at300	6.573
yield_E2.S08.rainfed	2.384
yield_E2.S08.best	6.566
yield_E2.S08.at300	5.853
yield_E2.S09.rainfed	2.773
yield_E2.S09.best	6.622
yield_E2.S09.at300	6.584
yield_E2.S10.rainfed	2.75
yield_E2.S10.best	6.59
yield_E2.S10.at300	6.575
collapse_raw_h	0.5407
collapse_scaled	0.3167
spearman_scaled	-0.8339
E10_runs	954
E10_runs_irrigated	744
fate_E10.S01.dT	0.38
fate_E10.S01.dE	0.323
fate_E10.S01.dO	0.186
fate_E10.S01.dD	0.08
fate_E10.S01.dS	0.03
fate_E10.S02.dT	0.232
fate_E10.S02.dE	0.186
fate_E10.S02.dO	0.261
fate_E10.S02.dD	0.316
fate_E10.S02.dS	0.004
fate_E10.S05.dT	0.638
fate_E10.S05.dE	0.113
fate_E10.S05.dO	0.137
fate_E10.S05.dD	0.075
fate_E10.S05.dS	0.038
fate_E10.S06.dT	0.409
fate_E10.S06.dE	0.22
fate_E10.S06.dO	0.174
fate_E10.S06.dD	0.152
fate_E10.S06.dS	0.045
fate_E10.S07.dT	0.352
fate_E10.S07.dE	0.295
fate_E10.S07.dO	0.187
fate_E10.S07.dD	0.148
fate_E10.S07.dS	0.018
fate_E10.S08.dT	0.459
fate_E10.S08.dE	0.184
fate_E10.S08.dO	0.164
fate_E10.S08.dD	0.147
fate_E10.S08.dS	0.045

quantity	value
fate_E10.S09.dT	0.363
fate_E10.S09.dE	0.287
fate_E10.S09.dO	0.181
fate_E10.S09.dD	0.153
fate_E10.S09.dS	0.016
fate_E10.S10.dT	0.341
fate_E10.S10.dE	0.292
fate_E10.S10.dO	0.177
fate_E10.S10.dD	0.147
fate_E10.S10.dS	0.044
fate_E10_all.dT	0.377
fate_E10_all.dE	0.252
fate_E10_all.dO	0.186
fate_E10_all.dD	0.156
fate_E10_all.dS	0.029
dose.S01.inf	0.109
dose.S01.dD	4.425
dose.S01.r2	0.949
dose.S01.dose_maxT	20
dose.S01.dose_best	2
dose.S02.inf	0.445
dose.S02.dD	3.234
dose.S02.r2	0.955
dose.S02.dose_maxT	2
dose.S02.dose_best	2
dose.S05.inf	0.102
dose.S05.dD	5.21
dose.S05.r2	0.658
dose.S05.dose_maxT	15
dose.S05.dose_best	15
dose.S06.inf	0.197
dose.S06.dD	3.887
dose.S06.r2	0.931
dose.S06.dose_maxT	15
dose.S06.dose_best	2
dose.S07.inf	0.194
dose.S07.dD	4.106
dose.S07.r2	0.929
dose.S07.dose_maxT	15
dose.S07.dose_best	2
dose.S08.inf	0.19
dose.S08.dD	3.85
dose.S08.r2	0.945
dose.S08.dose_maxT	20
dose.S08.dose_best	8
dose.S09.inf	0.204
dose.S09.dD	4.388
dose.S09.r2	0.926

quantity	value
dose.S09.dose_maxT	15
dose.S09.dose_best	2
dose.S10.inf	0.183
dose.S10.dD	3.394
dose.S10.r2	0.923
dose.S10.dose_maxT	15
dose.S10.dose_best	2
dose_predictors.n.rho_dD	0.07143
dose_predictors.n.rho_inf	0.02381
dose_predictors.taw.rho_dD	-0.2381
dose_predictors.taw.rho_inf	0.1905
dose_predictors.se100.rho_dD	-0.1667
dose_predictors.se100.rho_inf	0.1905
dose_predictors.dep300.rho_dD	0.119
dose_predictors.dep300.rho_inf	-0.04762
dose_predictors.store.rho_dD	0.1667
dose_predictors.store.rho_inf	0.1905
dose_table.S01.dD	4.425
dose_table.S01.inf	0.1094
dose_table.S01.n	1.353
dose_table.S01.taw	130
dose_table.S01.se100	0.4445
dose_table.S01.dep300	0.3794
dose_table.S01.store	14.8
dose_table.S02.dD	3.234
dose_table.S02.inf	0.4448
dose_table.S02.n	1.6
dose_table.S02.taw	105
dose_table.S02.se100	0.3563
dose_table.S02.dep300	0.4964
dose_table.S02.store	15.64
dose_table.S05.dD	5.21
dose_table.S05.inf	0.1017
dose_table.S05.n	1.89
dose_table.S05.taw	95
dose_table.S05.se100	0.1647
dose_table.S05.dep300	0.6274
dose_table.S05.store	17.88
dose_table.S06.dD	3.887
dose_table.S06.inf	0.1971
dose_table.S06.n	1.56
dose_table.S06.taw	150
dose_table.S06.se100	0.4663
dose_table.S06.dep300	0.4678
dose_table.S06.store	21.05
dose_table.S07.dD	4.106
dose_table.S07.inf	0.194
dose_table.S07.n	1.41

quantity	value
dose_table.S07.taw	220
dose_table.S07.se100	0.6859
dose_table.S07.dep300	0.3665
dose_table.S07.store	24.19
dose_table.S08.dD	3.85
dose_table.S08.inf	0.19
dose_table.S08.n	1.48
dose_table.S08.taw	160
dose_table.S08.se100	0.417

S7. Where every table and figure comes from

Item	How it is made	Script(s)	Input and output
Figure 1	Schematic drawn by the script with numbers from the results	Manuscripts/reproducibility/analysis/p4_concept.py	Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig1_concept.pdf; numbers Manuscripts/results/p4_numbers.json
Table 1	Table written by the script (tex/data_table.tex)	Manuscripts/reproducibility/analysis/data_overview.py	Data/Comprehensive_Summary/00_summary_all_simulations
Figure 2	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/data_overview.py	Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/data/fig_data_overview.pdf
Figure 3	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p4_reach.py	Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig2_curves_reach.pdf; data Manuscripts/results/p4_tableA1_reach.csv
Table 2	Table written by the script	Manuscripts/reproducibility/analysis/p4_reach.py	Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/results/p4_table2_laws.csv (sheet R_p4_table2_laws)
Figure 4	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p4_reach.py	Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig3_trigger_reach.pdf; data Manuscripts/results/p4_fig3_trigger_runs.csv, p4_tableA1_reach.csv
Figure 5	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p4_reach.py	Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig4_marginal_fate.pdf; data Manuscripts/results/p4_fig4a_fate_by_soil.csv, p4_fig4b_fate_by_method.csv
Figure 6	Figure drawn by the script from the input data	Manuscripts/reproducibility/analysis/p4_reach.py	Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig5_dose.pdf; data Manuscripts/results/p4_fig5_dose.csv
Figure 7	Screen capture of SWAFL Desktop 1.1.0 (not regenerated)	-	Manuscripts/figures/shots_jpg/real_trigger_law.jpg; zaria_matric_potential.jpg; wb_step3_experiments.jpg
Table 3	Descriptive table (claims and the evidence cited in the text); no computation	-	

Item	How it is made	Script(s)	Input and output
Table 4	Table written by the script	Manuscripts/reproducibility/analysis/p4_reach.py	Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/results/p4_tableA1_reach.csv (sheet R_p4_tableA1_reach)
Table 5	Descriptive table (this map in short); no computation	-	

References

Richard G. Allen, Luis S. Pereira, Dirk Raes, and Martin Smith. Crop evapotranspiration: Guidelines for computing crop water requirements. Technical Report FAO Irrigation and Drainage Paper 56, FAO, Rome, 1998. URL https://www.fao.org/4/x0490e/x0490e00.htm.

Shmuel Assouline and Dani Or. The concept of field capacity revisited: defining intrinsic static and dynamic criteria for soil internal drainage dynamics. *Water Resources Research*, 50, 2014. doi: 10.1002/2014WR015475.

A. Carminati and M. Javaux. Soil rather than xylem vulnerability controls stomatal response to drought. *Trends in Plant Science*, 25(9):868–880, 2020.

E. Diamantopoulos and W. Durner. Dynamic nonequilibrium of water flow in porous media: a review. *Vadose Zone Journal*, 11(3):vzj2011.0197, 2012. doi: 10.2136/vzj2011.0197.

E. Diamantopoulos, S. C. Iden, and W. Durner. Inverse modeling of dynamic nonequilibrium in water flow with an effective approach. *Water Resources Research*, 48(3), 2012.

S. Elmaloglou and E. Diamantopoulos. Wetting front advance patterns and water losses by deep percolation under the root zone as influenced by pulsed drip irrigation. *Agricultural Water Management*, 90(1–2): 160–163, 2007.

S. Elmaloglou and E. Diamantopoulos. The effect of intermittent water application by surface point sources on the soil moisture dynamics and on deep percolation under the root zone. *Computers and Electronics in Agriculture*, 62(2):266–275, 2008.

S. Elmaloglou and E. Diamantopoulos. Effects of hysteresis on redistribution of soil moisture and deep percolation at continuous and pulse drip irrigation. *Agricultural Water Management*, 96(3):533–538, 2009.

W. R. Gardner. Dynamic aspects of water availability to plants. *Soil Science*, 89(2):63–73, 1960.

Yechezkel Mualem. A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research*, 12:513–522, 1976. doi: 10.1029/WR012i003p00513.

P. J. Ross and K. R. J. Smettem. A simple treatment of physical nonequilibrium water flow in soils. *Soil Science Society of America Journal*, 64:1926–1930, 2000.

Navin K. C. Twarakavi, Masaru Sakai, and Jiří Šimůnek. An objective analysis of the dynamic nature of field capacity. *Water Resources Research*, 45:W10410, 2009. doi: 10.1029/2009WR007944.

Martinus Th. van Genuchten. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44:892–898, 1980. doi: 10.2136/sssaj1980.03615995004400050002x.